
ARTICLE

Two-Dimensional Signal Decomposition for Feature Reduction and Deep Learning-Based Object Detection

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ABSTRACT: Object detection is one of the important and widely used methods for the identification of target objects in a specific image and for the determination of the position and category for achieving machine vision's understanding. A lot of approaches are proposed in the literature for obtaining a solution to this problem. Best results have been obtained by using deep learning and computer vision-based methods. However, most of the existing techniques perform under expectation, particularly in detecting the small and dense objects and in the detection of objects which has random transformations in geometric measures. The sparse representation methods often fail to perform particularly in cases where the rate of recognition is higher. The problem of object detection can be represented in three stages, namely the representation, dimension reduction, and object detection. Initially, a dictionary is used for the representation of samples undergoing test, followed by a sparse representation. Secondly, in order to resolve the issue of high dimensionality of the features, Bi-Dimensional Empirical Mode Decomposition (BEMD) is introduced for dimension reduction. Thirdly, an architecture called AlexNET is proposed for getting the multi-scaled feature and to add a convolutional structure for the detection of dictionaries for the identification of objects, which is used as a guideline for the selection of architecture for achieving performance measures such as speed, memory, and accuracy. Lastly, the visualization of images from the COCO dataset, which shows side-to-side comparison with the existing methods which lies on the "optimal front" of the speed vs. accuracy plot. The experimental results are compared with the state-of-the-art techniques in various performance metrics, with emphasis on speed and accuracy of the latest methods of object detection.

KEYWORDS: Object Detectors, Bi-dimensional Empirical Mode Decomposition (BEMD), deep learning, AlexNet, Sparse Reduction, and Common Objects in Context (COCO) dataset.

1 Introduction

Object detection is a basic research issue in the computer vision domain, and numerous methods are available in the literature for addressing the challenges. The importance of object detection is for the identification of one or more targets image/video. It includes a large range of methods such as image processing, recognition of patterns, and ML. The applications of these are widely adopted, such as traffic control and prevention of accidents [1], to issue warnings in factories, military-related monitoring, and in advanced human-system interaction [2,3]. As the application domains are mostly



in the use cases of multi-target scenarios, the relationship between the accuracy and cost has become difficult to balance.

The process of object detection is conventionally established through manual extraction of the models of features, where commonly identified features are represented through Histogram of Oriented Gradient (HOG), SIFT, Haar-based features, and other classical techniques that rely on grayscale. SVM-based methods or Adaboost techniques are normally used for classification to obtain target data. These classical methods of extracting models were only capable of determining low-level data on the features, such as information on contour and textures, and have limitations for detecting multiple levels of targets under complex scenarios, owing to poor performance.

Numerous studies have been made in the recent past on the issues of object detection owing to the usage of CNNs. These are ML models that come under Deep learning. In order to perform image recognition, the CNNs can make use of the image information as inputs without pre-processing and additional feature extractions. Hence, Conventional Neural Networks (CNNs) are very much suitable for the extraction of features in images, which include shapes, textures, and topology. Advanced methods such as Fast R-CNN [4], R-FCN [5], SSD [6], multi-box [7], and YOLOs [8] are presently good for deployment in consumer-based products, which normally run in mobile devices.

Server-based production techniques, which are used in web applications such as Google or Facebook, have more importance to accuracy but are still subject to throughput. Advanced methods such as the COCO challenge [9] are more optimized in terms of accuracy, often rely on models that ensembles multi-crops-based techniques, which are very slow to be used practically. Only a few researches [5,7,8] addresses the issue of running time in a detailed manner. Further, it is also identified that these methods are good in achieving frame-rate but do not provide a clear picture of the speed and accuracy that relies on other factors such as the extractor, input size, and image, etc.

It is highly difficult for the researchers to come to a conclusion on which architecture shall be used for the application, and also the standard metrics of accuracy. The average precision (mAP) does not speak much here, as the real-time deployment of computer vision, running time, and usage of memory are also vital. It is identified that the existing methods perform poorly in order to detect small and dense objects, and even more difficult to identify the objects that undergo frequent geometric transformations. With an aim of high-dimensional problems including deep features, the Bi-Dimensional Empirical Mode of Decomposition (BEMD) is introduced for reducing the dimensions. It is used for serving as a guideline for identifying a suitable architecture that achieves higher speed and less memory usage with high accuracy in a specific application. Finally, AlexNet is introduced for object detection, and the architecture is proposed for obtaining multi-scaled features and which adds deformable structures. Experiments are carried out to trace the speed vs accuracy curve for various detection systems, which vary in meta-architecture, feature extraction, and image resolution. Etc.

2 Literature Review

Bell et al [10] proposed Inside-Outside Net (ION) for detecting objects that exploit the information both in and out of the regions of interest. The context information, which lies outside the regions of interest, is integrated with the aid of spatial and recurrent neural networks. On the inner side, a skip pooling technique is used for the evaluation of design spaces, which provides readers with an overview of which methods are most important. The proposed method improves the most used PASCAL-based VOC method of detecting objects from 73.9 to 77.9% mAP. The proposed model has shown an improvement of 19.6% to 33.1% in mAP in the challenging MS common object in context (COCO). The proposed method confirms that the usage of ION plays a vital role in the multi-scale representation and improves the detection of small objects.

Erhan et al [11] proposed the Deep Convolution Neural Network (DCNNs), which achieves state-of-the-art performance in numerous image recognition including the ImageNet Visual Challenge of Recognition (ILSVRC-2012). The proposed model is based on the localization, which uses a single bounding box and a score for the confidence of every object in the image. Such models capture contexts of the whole image around the required objects, but cannot handle the issue of multiple instances of the same object. The proposed method uses a saliency-based neural network for object detection that predicts the set of bounding boxes termed as class-agnostics with scores for every box, and the corresponding likelihoods are calculated. The proposed method inherits a number for each variable of instances and for each class, a generalization is allowed in a high-level order of the network. The proposed method has the advantage of having a grid in regular grid that predicts the boxes that are written as predictors on the images, which has shared parameters.

Fu et al [12] proposed the Single Shot Multibox (SSD) method for the introduction of additional contexts in the modern methods for detecting objects. In order to achieve the same, the authors have combined a classifier with a rapid detection framework called SSD. The proposed SSD with a de-convolution layer is introduced with additional contexts for the detection of objects and to improve accuracy, especially for the small objects, which calls the result system – DSSD for de-convolution single shot-based detector. The scenario of using these contributions can be easily described in a high-level manner, but a traditional method does not give success. The experimental results show that a potential way forward is achieved in terms of the detection parameters of PASCAL, VOC, and COCO.

Kim et al [13] investigated how accuracy can be achieved at the maximum level in the case of multi-category object detection, while minimizing the cost. The proposed method adapts and combines the recent methodologies. The technique also follows the common pipeline of CNN-based feature extraction and ROI classification that aligns with the recent techniques such as truncated SVDs. The principle of design adapted here is the “less channeled and more layered” approach with adaptation of building blocks, which includes concatenated ReLU and Hypernet. The proposed framework is deep as well as thin, which is trained with the aid of batch-based normalization and with residual-enabled connections, where the rate of scheduling is based on plateau detection.

Lin et al [14] proposed the technique that uses a multi-scale and parametric hierarchy in a deep convolution network for constructing a features’ pyramid that has marginal additional cost. The top-down approach with laterally biased connections is developed to build high-level semantics-based maps in all scales. The proposed Feature Pyramid Network (FPN) proved significantly better when compared with generic methods of feature extraction. The technique uses a fundamental R-CNN system that achieves state-of-the-art results on the COCO models without whistles, which includes those from the 2016 challenge winners. Finally, the technique, which can run at 5 frames per second on a single GPU, has also been developed, bringing out practical and more accurate solutions for object detection.

Shrivatsav et al [15] introduced a simple but powerful technique called the Online Hard Example Mining (OHEM) for the training of ConvNet detection. The authors are motivated by the dataset, which had higher count of easy objects and a small count of hard objects. OHEM uses a simple algorithm that eliminates the heuristics and extra parameters in common use. The proposed method consistently increases the performance on standards such as PASCAL VOC 2007 and 2012. The effectiveness of the proposed method increases as the size of the data increases and yields better results in the MS COCO dataset. Moreover, these are combined with the advances of OHEM, which leads to better results with 78.9% mAP.

Huang et al [16] conducted research on different types of trade accuracy for ensuring speed and the usage of memory in modern methods of CNN-based object detection systems. A range of successful techniques has been introduced in the recent past, but comparisons such as apples-to-

apples were difficult. A unified mode of implementation on the fast region-based CNN and SSD methods is adopted by making use of alternate feature extraction and other parameters, such as the size of the image within all of the meta-architecture. At one end of the spectrum, where the speed as well as the memory are vital, a common detector is introduced that can achieve all the speeds required in real-time in a mobile device. As far as the accuracy is concerned, a detector is introduced that can achieve better performance based on the COCO detections. Wei et al [17] introduced an image recognition approach based on DNN with adaptive and weight-based Joint Sparse Representation (D-AJSR). The proposed model is a data-lightweight framework that can classify and recognize objects well within a few training samples. In the proposed method, CNN is used for the extraction of deep features on the training samples as well as the testing samples. Then, the AD-WJSR method is adopted for sparse representation, where the Eigen vectors are again reconstructed through the calculation of each of the eigen vectors. Aiming at the problem of high dimension, the PCA is implemented for dimension reduction. At last, these models are combined with the sparse model, the common features, and the private features of these images, which are extracted through the samples, for constructing a dictionary. Sparse-based classifiers are used for object detection. The experimental results show that the proposed method is superior to the existing methods.

Lu et al [18] introduced a method for object detection which are based on an improvised R-CNN technique. The improved method can easily identify the regions where the objects belong to and are scattered in different complex appliances more quickly and in an automated manner. The feature-enhanced framework named the deeper proposal network (RPN) is an enhancement model for the effective extraction of features in an object in a kitchen appliance scene. The U-shape network is then restructured with the aid of a series of enhancement modules of the feature. The experiments performed best in object detection and have attained a mean average precision of 89%. The experiments also prove that the proposed method achieves more detection accuracy than the other modern methods of object detection. The proposed method also works well when applied to the recognition of texts. Cao et al [19] made an analysis of the mainstream-based object detection methods and a multi-scaled deformable object detection framework, which are proposed for dealing with the issues faced by the present methods. The research analysis demonstrates that the strong performance is on par with or even better than the state of the methods. Deep methods are introduced for obtaining multi-level features and to add DCS structures for overcoming the geometric transformations. Fusion of multi-scaled and multi-level features is generated using sampling for the implementation of the final recognition of objects and regression. The experiments prove that the suggested frameworks can improve the efficiency in detecting the small objects that have geometrical deformation, which shows a consistent improvement in the trade-offs between speed and accuracy.

3 Proposed Methodology

In the initial stage, a dictionary is made for representing the samples taken for testing, and then the sparse representation coefficients are used. Next, in order to resolve the issue of high-dimensionality, the Bi-Dimensional Empirical Mode Decomposition (BEMD) is introduced. Thirdly, the architecture named Alexnet is implemented for obtaining the features that are multi-scaled in nature, and to which the deformable CNNs are added for detecting the dictionary for object identification. Primarily, the four recent meta-architectures, the SSD, R-CNN, R-FCN, and AlexNet, are proposed.

3.1 Sparse Representation

This framework for object detection is then gradually implemented for other classifications of images and object recognition. If there are N images in training, $X = [x_1, x_2, \dots, x_n]$ ($x_i \in \mathbb{R}^m$, general $m \ll n$), that belong to K number of classes, then all the training data shall be given as,

$$W = [W_1, \dots, W_k] \in \mathbb{R}^{m \times n} \quad (1)$$

Here, $W_i = [v_{i,1}, v_{i,2}, \dots, v_{i,n_i}]$, $v_{i,j}$ denotes the j^{th} number of samples in the i^{th} class, and n_i denotes the number of samples in the i^{th} class. According to the theory of Sparsed Representation, SR, fresh test samples such as $y \in \mathbb{R}^m$ of the class “ i ” shall be expressed linearly by the sample taken for training, and the same is represented as W_i as in [20]:

$$y = \alpha_{i,1}v_{i,1} + \alpha_{i,2}v_{i,2} + \dots + \alpha_{i,n_i}v_{i,n_i} \quad (2)$$

where $\alpha_{i,j} \in \mathbb{R}$ is the sparse representation coefficient of y , $j \in 1, 2, \dots, n_i$. Without considering the noise, equation (2) can be written as equation (3),

$$y = Wx \quad (3)$$

where $x = [0, \dots, 0, \alpha_{i,1}, \alpha_{i,2}, \dots, \alpha_{i,n_i}, \dots, 0]^T \in \mathbb{R}^n$. To get the sparsest x , object detectors need to solve the following l_1 minimization problem by equation (4),

$$x' = \arg \min \|x\|_1 \text{ s.t } Wx = y \quad (4)$$

Here, x denotes the coefficient vectors and $\|x\|_1 = \sum_i |x_i|$ is in l_1 norm. For each of the classes, a mapping function is introduced, such as $\delta_i(x') = [0, \dots, 0, x_i, 0, \dots, 0]$ for representing the non-zero elements which are selected in the corresponding vector of the i^{th} class, the samples taken for test are then reconstructed using the coefficient of sparsity as $y' = W\delta_i(x')$. Hence, y is determined for classification in class(i) through the minimum residual through equation (5),

$$\text{class}(i) = \arg \min_i \|y - W\delta_i(x')\|_2 \quad (5)$$

In the above equation, $\|\cdot\|_2$ is l_2 norm and $i = 1, 2, \dots, k$. In case of sparse representations, owing to the correlation between different images, the related images can be used as a single set where every image shall be represented in terms of a combination of public and private means of features in a specific sparse representation. The public features are a part of all images in one set, whereas the private features are the characteristic part and are unique to every image. Hence, the j^{th} image shall be represented by means of public features and can ascertain its own features, which are private and are represented as,

$$y_j = z_c + z_j, j \in \{1, \dots, J\} \quad (6)$$

In the above equation, z_c depicts the features that are classified as public and z_j denotes the features that are private to the j^{th} image. Assuming such that the images are divided into K number of classes and as there are J images for training in every training image in every category. If a particular image is represented in terms of a single-dimensional vector, the image on the i^{th} object can be given in terms denoted as $y_i \in [y_{i,1}, y_{i,2}, \dots, y_{i,j}]^T$. As far as the sparse representations are considered, the j^{th} image of the i^{th} class shall be represented in the equation as follows,

$$y_{i,j} = z_i^c + z_{i,j}^i, j \in \{1, \dots, J\} \quad (7)$$

In the above equation, z_i^c denotes the features that are public on the images of the i^{th} number of object and $z_{i,j}^i$ denotes the own private features. In case of $\Psi \in \mathbb{R}^{N \times N}$ is orthonormal, which can represent the training image, then the equation (7) shall be represented as in (8),

$$\theta_{i,j} = \Psi z_i^c + \Psi z_{i,j}^i = \theta_i^c + \theta_{i,j}^i \quad (8)$$

Here, the $\theta_{i,j} \in \Psi y_{i,j}$ defines the representation of $y_{i,j}$ based on transformation and Ψ and $\theta_i^c = \Psi z_i^c$ and $\theta_{i,j}^i = \Psi z_{i,j}^i$ represents the parts in terms of public and private denoted by Ψ , respectively. Multiplied by Ψ^T , on both the sides, then (8) is reduced to (9),

$$\Psi^T \theta_{i,j} = \Psi^T \Psi z_i^c + \Psi^T \Psi z_{i,j}^i = \Psi^T \theta_i^c + \Psi^T \theta_{i,j}^i = z_i^c + z_{i,j}^i \quad (9)$$

Combined with equation (7), $y_{i,j} = \Psi^T \theta_i^c + \Psi^T \theta_{i,j}^i$, so the joint representation of the image can be expressed by equation (10),

$$\begin{bmatrix} y_{i,1} \\ y_{i,2} \\ \vdots \\ y_{i,j} \end{bmatrix} = \begin{bmatrix} \Psi^T & \Psi^T & 0 & \dots & 0 \\ \Psi^T & 0 & \Psi^T & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \Psi^T & 0 & 0 & \dots & \Psi^T \end{bmatrix} \cdot \begin{bmatrix} \theta_i^c \\ \theta_{i,1}^i \\ \theta_{i,2}^i \\ \vdots \\ \theta_{i,j}^i \end{bmatrix} \quad (10)$$

Equation (10) can be simplified to equation (11),

$$y_i = \tilde{\Psi} W_i \quad (11)$$

Here, $y_i = [x_{i,1}, x_{i,2}, \dots, x_{i,j}]^T$ and $W_i = [\theta_i^c, \theta_{i,1}^i, \theta_{i,2}^i, \dots, \theta_{i,j}^i]^T$. $\tilde{\Psi} = [A, B]$ are the complete dictionary, which consists of a couple of parts; $A = [\Psi^T \Psi^T \dots \Psi^T]^T$ and $B = \text{diag}(A)$. W_i preserves the discriminated information, and the same representation in sparse shall be obtained through solving the (l_1) minimization of the following equation (12),

$$W_i = \arg \min \|W_i\|_1 \text{ s.t. } y_i = \tilde{\Psi} W_i \quad (12)$$

After obtaining W_i , the public features of all images in object i , and the private features of each image can be obtained in the Ψ field according to the inverse transformation,

$$z_i^c = \Psi^T \theta_i^c \quad (13)$$

$$z_{i,j}^i = \Psi^T \theta_{i,j}^i \quad (14)$$

All the features that are public form a joint feature D , and the same is represented as in (15),

$$D = [z_1^c, z_2^c, \dots, z_k^c, z_{1,1}^1, \dots, z_{1,j}^1, z_{2,1}^2, \dots, z_{2,j}^2, \dots, z_{K,1}^k, \dots, z_{K,j}^k] \quad (15)$$

3.2 Bi-dimensional Empirical Mode Decomposition (BEMD)

The process of iteration and shifting of Bi-Dimensional Empirical Mode (BEMD) is the same as that of EMD. The method is a time-bound method of analysis, especially suited for non-linear and non-stationary data. The important idea is to find an intrinsic multi-scale variation of the input image. Du et al [21], intrinsic functions are expressed through the Equation (16) as follows,

$$X(t) = \sum_{i=1}^n \text{IMF}_i + R_n \quad (16)$$

Here, the $X(t)$ indicates the image taken as input, and the R_N is the residue, which is decomposed into Intrinsic Mode Function (IMFs) and residue. The images can be considered as a 2D matrix $f(x,y)$.

3.2.1 Definition of Extrema Point

Given a one-dimensional input, it is easy to define the required peaks and the valleys of an image as the extremes. The identification of these extremes in a two-dimensional image is then performed by searching for the same in octa directions individually along the rows and columns, including diagonals, in a matrix of pixels, which are tabulated. If a given point is taken as a peak or as a valley in all directions, then the same is called a two-dimensional extreme. As the BEMD involves more shifting and iteration, the time consumption is higher for identifying the extrema. Initially, the extremes along the columns are identified, and then the rows that have the extrema values of the corresponding columns are found. This sequence helps in saving time. A condition is termed exceptional when there is a flat peak in the image. In this scenario, the middle point is only considered.

3.2.2 Interpolation and Smoothness

After obtaining the extreme points, the next step aims to get the maximum and the minimum surface envelopes through the smoothness and through the interpolation. Triangulation is used in the cubic spline interpolation of the traditional EMD technique. If the point is seen outside the convex hull, the triangulation fails on the point, which results in its infinity, and may also cause the boundary effect of dimension reduction. However, the function is a distance-based interpolation technique where N points of pixels are pointed to a full $N*N$ matrix that shall be generated. A system of equations is then solved by making use of the matrix. Hence, the function tends to be quieter and slow down even when applied to problems of even size. In order to tackle the above issues, the Gridoff tool is used for the extrapolation of the outside convex hull and for generating a smooth surface envelope for the reduction of the actual input matrix of the image.

3.2.3 Stopping Criteria in the Iteration Process

In most conditions, the value obtained for IMFs that use the BEMD cannot satisfy the conditions of the IMF's definition. The global solution is to be proposed for judging the convergence. Additional iterative decomposition shall not provide excess information when the criteria for convergence are very small. As an alternative, one can also make use of the methods with a stopping criterion. The criteria for SD are denoted as,

$$SD = \sum_{i=0}^x \sum_{j=0}^y \left[\frac{|H_{i,(j-1)}(x,y) - H_{ij}(x,y)|^2}{H_{i,(j-1)}^2(x,y)} \right] < r \quad (17)$$

Where SD is a stopping criterion. r is defined as a constant that is pre-defined in the range of 0.2 to 0.3. The details of the decomposition methods (BEMD) are given as,

1. To find all the local minimum and maximum of the function $f(x,y)$ to reduce the dimension of the original matrix image.
2. To perform the interpolation of the surface that uses the extrema from step 1 to obtain the maximum envelope surface $E_{max}(x,y)$ with the minimum value of the envelope surface $E_{min}(x,y)$. Here, the extension of the image is necessary for avoiding the invalid interpolations caused by boundary conditions.
3. To calculate the mean envelope surface $Avg(x,y)$, which is based on the maximum envelope surface and the minimum envelope surface.
4. H_{ij} is then obtained through the subtraction of the average envelope from the basic signal $f(x,y)$, where; H_{ij} indicates the j^{th} iterations of the i^{th} process of shifting.

5. To confirm whether the criteria for stopping are satisfied. If not, then the H_{ij} is defined as $f(x, y)$, and to repeat the steps from 1 to 4. If the condition for stopping is already satisfied, it is possible to obtain the IMF_i as,

$$IMF_i(x, y) = f(x, y) - H_{ij}(x, y) \quad (18)$$

This is termed as complete process of shifting. The steps through 1-5 are then repeated till the threshold becomes a constant and which indicates that the BEMD process of shifting is complete.

3.3 Object detection

The anchor methodology is intended to minimize the losses in terms of combined regressions and classification. For every anchor, an initial and best-matching truth box b is proposed. If there is a match found, it is termed as “positive anchor” and assigned as (1) in a class label $y_a \in \{1, \dots, K\}$ and (2) denotes the vector encoding of the box b corresponding to an anchor ‘a’ and set to the class label as $y_a = 0$. If an anchor is a predicted box that has the encoding of $f_{loc}(I; a; \theta)$ which corresponds to the class $f_{cls}(I; a; \theta)$, where I denotes the image and θ denotes the model’s parameter. In this context, the loss for ‘a’ is then measured as the weighted sum of the location-based losses and the classification-based losses, and the same is represented as,

$$\mathcal{L}(a, \mathcal{I}; \theta) = \alpha \cdot 1[a \text{ is positive}] \cdot \ell_{loc}(\phi(b_a, a) - f_{loc}(\mathcal{I}; a, \theta)) + \beta \cdot \ell_{cls}(y_a, f_{cls}(\mathcal{I}; a, \theta)) \quad (19)$$

Here, α, β denote the weights of the balancing localizations as well as the losses due to classification. In order to train the model, Equation (19) is leveraged over the anchors and also minimized in terms of parameter θ . In the case of multibox[11], these “box priors” are generated by the clustering of truth boxes of the dataset. In advanced techniques, these anchors are then generated through the tiling of a collection of various boxes in different aspect ratios and in different scales regularly across the given image. One advantage of having such an anchor is that these predictors shall be written as tiled predictions of the image that share parameters and are reminiscent of conventional sliding window techniques. The fast R-CNN[4] and the latest multibox[6] are the ones to adopt this technique in the preliminary experiments.

3.3.1 Single Shot Detector (SSD)

This refers to the broad architecture which uses a single feed-forward network for direct prediction of classes and the offsets of anchors that does not require a second stage per-proposal of classification operation [22]. Under these definitions, the SSD architecture has been explored in numerous precursors as mentioned in [22]. Both the Multibox and the RPN stage of Faster CNN [6] can use this approach for predicting the class-agnostic boxes for predicting the final class labels [12].

3.3.2 Faster R-CNN

In the case of the faster R-CNN settings, the detection is expected to be in a couple of stages. In the initial stage, the region proposals are called using the RPNs, and the images are processed using a feature extractor, and the features are selected at an intermediate level, such as in conv5 are used for the prediction of class-agnostic proposals. The loss functions are taken in the form as in equation(19), which uses the anchor grids. In the later stage, these proposals are then used for the cropping of features in the same intermediate maps that are subsequently fed into the remains of the extractor (such as in fc7) for predicting the class and class-specific refinement of every proposal. The loss functions for these stages also take the form of equation(19) by using the proposals that are generated from the RPN of anchors. It is also to be noted that the crop proposals are obtained directly from the image and are re-run using the feature extractions, which can be duplicated in computations. However, there is a part of the computations that must be run only once in a region,

which makes the run time dependent on the number of independent regions that are proposed by RPN.

3.3.3 Region-based Fully Convolutional Networks (R-FCN)

The faster R-CNN is considered in terms of magnitude, which is faster than the R-CNN, the facts that these are region-specific components that must be applied several times on an image. Dai et al[15] have proposed the R-FCN methods, which are like the Faster R-CNN, but here, instead of features being cropped in the same layer, the crops are instead taken from the final layers of feature prediction. This kind of approach, by pushing the cropping into the last layers, is then taken from the features before the prediction. This kind of pushing the cropping into the last layer actually minimizes the volume of per-region computations that need to be performed. Dai et al[5] recommend that the object recognition task needs to localize the representations that respect the translation variance, and thus, which can propose the sensitive mechanisms that are used instead of more standard ROI operations[4] and the differentiated crop technique. They, in turn, show that the R-FCN model, which uses the Resnet 101, should achieve more accuracy for faster CNNs often than the running times.

3.3.4 AlexNet CNN deep learning architecture

In this work, the AlexNET architecture with CNN deep learning[22] is introduced for the detection of objects. The network is presented in a deeper way than the regular CNN with 5 convolution layers, which is followed by three kinds of pooling layers. A dropout of 0.5 percent can be applied to the fully connected layers to avoid the issue of overfitting. The proposed architecture is shown in Figure 1.

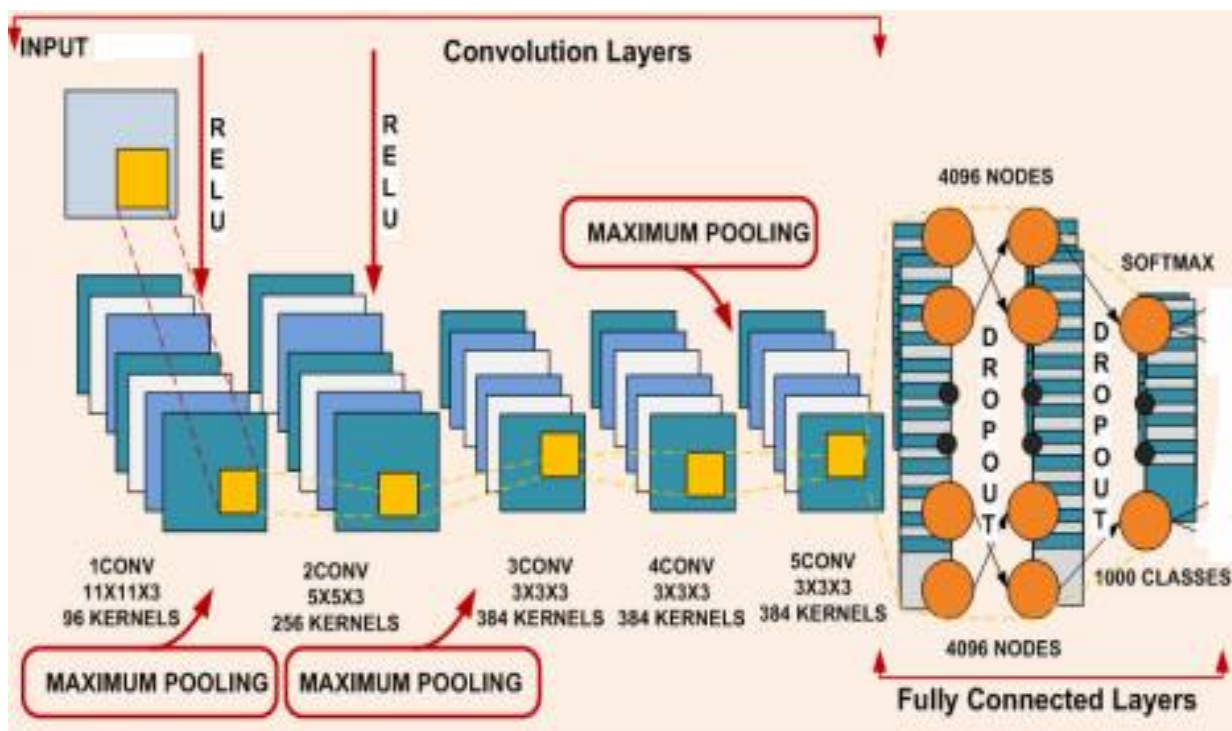


Figure 1: AlexNET convolution neural network architecture of the proposed framework

In this work, the input layer is presented as the pre-processing layer where the input images are sampled from the initial size to 227×227 in spatial resolution to reduce the cost of computation. The proposed system then used five CONV layers, which are then followed by the three POOL layers and a Rectifying Linear Unit (RELU).

Each of the convolutional layers generates a feature map. These maps are of the first, second, and the fifth layer, which are used in various combinations with the pooling layers of a 3*3 matrix and with the side of a 2*2 matrix. These frameworks are comprised of eight different layered architectures that have 4096 nodes. These generate the trainable feature sets in a feature extraction phenomenon of these layers. These are then subjected to Soft Max activation for the determination of classification probability used for the final classification output. These probabilities, which are in the softmax layer, can make categories up to 1000 various other classifiers.

3.3.4.1. Convolution Network Layer

This layer is significant in the deep learning context with neural networks, which generates feature maps that are confined to the classification. It has a kernel that can slide over the image, which is taken as input, and which generates an output called a feature map. In each location of the input, a matrix multiplication is performed, which is followed by the integration of the results. The feature map's output is then defined as,

$$N_x^r = \frac{N_x^{r-1} - L_x^r}{S_x^r} + 1, N_y^r = \frac{N_y^{r-1} - L_y^r}{S_y^r} + 1 \quad (20)$$

Here, N_x, N_y denotes the width as well as the height of the expected output of the feature map in the final layer, and L_x, L_y denote the size of the kernel S_x, S_y which depicts the total count of the pixels that are skipped through the kernel in a horizontal and vertical direction with index r that indicates the corresponding layer, i.e, $r=1$. Convolution is then applied to the input, and the kernel is used to get the output, which is defined as,

$$X_1(m, n) = (J * R)(m, n) \quad (21)$$

Here, $X_1(m, n)$ denote a 2D output of the feature map, which is obtained through the convolution of a 2D kernel R of the size (L_x, L_y) with the feature map denoted by J . The $*$ symbol is used for the representation of convolution between the J & R . The same is represented as,

$$X_1(m, n) = \sum_{p=-\frac{L_x}{2}}^{p=+\frac{L_x}{2}} \sum_{q=-\frac{L_y}{2}}^{q=+\frac{L_y}{2}} J(m-p, n-q) R(p, q) \quad (22)$$

In the proposed method, there are five layers of CONV along with RELU, and the normalization layers are used for extracting the maximum number of feature maps from the input stage for training the dataset to ensure maximum efficiency.

3.3.4.2. Rectified Linear Unit Layer

In the following stage, the activation of the RELU function is done for all the layers that are trainable for strengthening the network to make it non-linear. The proposed layer accounts for the non-linearity in all possible manners. It is then applied over the feature map, which has been generated from the CL. The use of **tanh(.)** and the RELU function makes the non-linear gradient to get saturate and results in terms of training time. The same is expressed as,

$$X_1(m, n) = \sum_{p=-\frac{L_x}{2}}^{p=+\frac{L_x}{2}} \sum_{q=-\frac{L_y}{2}}^{q=+\frac{L_y}{2}} J(m-p, n-q) R(p, q) \quad (23)$$

Values of the final feature maps are then obtained by applying the RELU, and the same is represented as,

$$X(m, n) = \begin{cases} 0, & \text{if } X_2(m, n) < 0 \\ X_2(m, n), & \text{if } X_2(m, n) \geq 0 \end{cases} \quad (24)$$

Here, the $x(m, n)$ is achieved through the transformation of negative inputs to zero and then returning the same on receipt of any positive values. The RELU layer is then considered to be the proposed framework as the deep neural networks are trained much faster and are intact with the RELU layers.

3.3.4.3. Maximum Pooling Layer

A layer for pooling is then included in the architecture proposed after the initial and final convolution, and then through the 5th CONV for decreasing the spatial size of every image to bring down the cost of computation of the framework proposed. The pooling operation simply picks the value that is maximum value for every slice on the image. The applications of the pooling layer, which is on the activation's output for down-sampling, are shown in Figure 2.

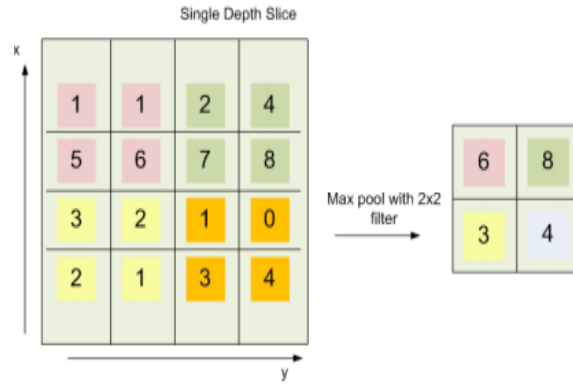


Figure 2: Maximum Pooling Layer

In the proposed method, the pooling is applied for using the maximum values against every slice, and the same brings better results with this mechanism.

3.3.4.4. Response Normalization Layer and the Softmax Activation

This is performed after two successful sessions to reduce the error while testing the proposed networks. The layer helps in normalizing the input layers inside the network all along with the inputs of the whole network, and the same is described as,

$$N_{e,f}^x = \frac{b_{e,f}^x}{\left(z + \alpha \sum_{j=\max(0, x-\frac{c}{2})}^{\min(T-1, x+\frac{c}{2})} (b_{e,f}^x)^2 \right)^Y} \quad (25)$$

Here, the $N_{e,f}^x$ represent the activity of normalization, consisting of $b_{e,f}^x$ number of neurons which are computed in position (e, f) within the use of Kernel K. T is the total range within the layers, z, c, and α are the given constants, which are hyperparameters of adjustable values through application of validation sets respectively. The soft-max is a classifier on top of the features extracted. After the

performance of five series of CNNs, the output is then fed into the soft-max layer for the multi-classification, which helps in determining the probabilities of the classification. These are, in turn, used by the final classification layer for classifying the input images into different classes.

3.3.4.5. Dropout Layer

This layer is then applied to the two layers, which are fully connected when the total count of the iterations gets doubled in the network to avoid the overfitting of data by substantially increasing the count of iterations by a factor of two, which makes the neurons dense. It then performs the average model along the neural network, which shows an efficient result in regularizing the data which are trained. The maximum number of layers that are used for pooling and the size of the kernel are processed in such a way that the output maps of the feature are down-sampled into a single pixel per map. Figure 3 then depicts the regularization of methods on layers that are fully connected prior to and after the dropout being applied.

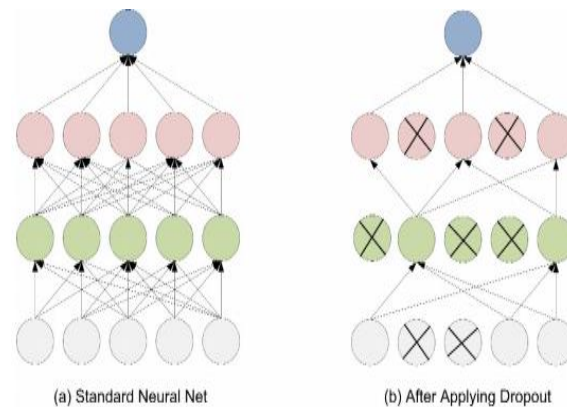


Figure 3: Fully Connected Layers (FC) before and after applying dropout

The layers that are fully connected also get connected to the output of the top layers in a single-dimensional vector. The upper layers are always fully connected to those of the output for a given class label, such that the high-level features are formed from the training data.

3.4. Matching

The targets of classification are determined for every anchor, which requires matching anchors for instances of ground truth. The common approaches include Greedy Bi-Partite (ex, which are based on Jaccard overlap) or many-one matching techniques where the Jaccard method is not used, but the matchings are kept discarded if the value of overlap is very low. Referring to these strategies, as bipartite or Argmax, the experiments use a throughput with given thresholds, which are suggested in the original work for every meta-architecture. Once the matching process is over, there is a typical sampling procedure which are designed for bringing the total count of positive anchors and the negative anchors in a desired ratio. In the experiments, recommended ratios are fixed for every meta-architecture.

4 Results and Discussion

This section analyses the data that has been collected during the training and benchmarking, and sweeps over the model configurations. Every model included here is a choice of meta-architectures with feature extractors stride (for Resnet and Inception), as well as the resolution of the input and number of proposals. These models were trained on some images scaled up to M

number of pixels in the shorter edges whereas in SSD, these images are always resized to a particular shape defined by $m \times m$. The evaluation models are then explored for down-scaled images to assess the accuracy and speed. In particular, these images are trained through different peaks of resolutions in each model. The experimental setup has $M=600$, and for high resolution and M is set to 300 for low resolution images. In either of the cases, it means that the SSD should at any time process only fewer pixels on average than a faster R-CNN, Alexnet, or R-FCN model that has all other variables which are held constant. The networks are then trained with the COCO data by making use of all training images as a substitute for image validation.

For every model, the time is measured in GPU, and the memory required, precision, and recall are calculated. The most important ones are defined below. True positive indicates the correct detection of a ground-truth bounding box and is represented as TP. A false positive is the instances of incorrect detection of a non-existent object in an existing object. It is represented as FP. A False negative is the instances where the ground-truth bounding boxes are undetected. It is vital to note that in the context of object detection, a TN does not apply, as there is an infinite count of bounding boxes that need not be detected in a certain image. Assessment of the detection mechanism is normally based on the parameters such as precision P , recall R , and the same is represented as,

$$P = \frac{TP}{TP+FP} = \frac{TP}{\text{All detections}} \quad (26)$$

$$R = \frac{TP}{TP+FN} = \frac{TP}{\text{All ground truths}} \quad (27)$$

$$F - \text{measure} = 2 \frac{P \cdot R}{P+R} \quad (28)$$

Precision is considered the ability of a model it identifies each relevant object. It is the percentile of the accurate positive count of predictions. Recall is the model's ability to understand and find all the cases that are relevant. It is calculated as the percentile of accurate predictions out of all the given truths. F-Measure denotes the harmonic mean of the precision and the recall together. In all the interpolations, it is seen that one can interpolate via all the points in such a way that are defined as follows,

$$AP_{all} = \sum_n (R_{n+1} - R_n) P_{interp}(R_{n+1}) \quad (29)$$

$$P_{interp}(R_{n+1}) = \max_{\tilde{R}: \tilde{R} \geq R_{n+1}} P(\tilde{R}) \quad (30)$$

Here, instead of using the precision that is observed by the few points, the AP can now be obtained through the interpolated precision at every level, which takes the maximum amount of precision for measuring the accuracy in object detection in all the classes of a given database. The mAP is simply the average AP over all classes [4], [7], which is described as,

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad (31)$$

With AP_i being the AP in the i^{th} class, and N is the total number of classes being evaluated. Figure 4(a-c) shows the input image, segmented image, and object detected of Air France. Figure 4(d-f) shows the input image, segmented image, and object detected of pizza.



(a) Air France image



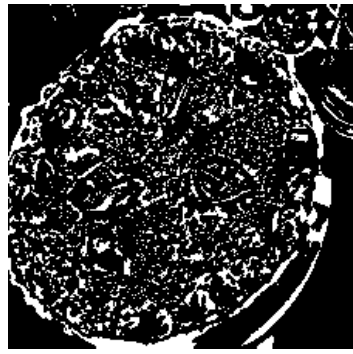
(b) Segmented image



(c) Object detected in the Air France



(d) Pizza image



(e) Segmented image



(f) Object detected pizza image

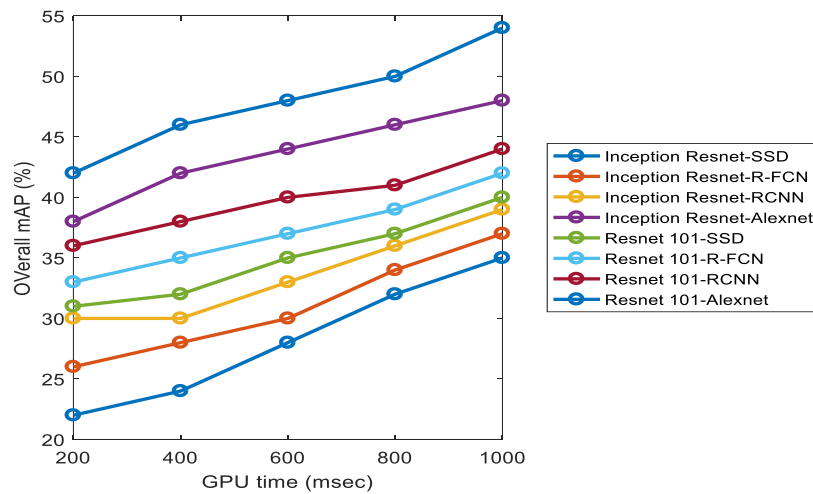
Figure 4: Image Results**Figure 5: Accuracy vs time comparison with respect to feature extractors**

Figure 5 shows some visualizations of the MAP in each of the classifiers along with feature extractors. It is seen from the results that it can be concluded that the proposed method, RCNN, and the AlexNet type of models are fast when compared to the R-FCN. In terms of accuracy, R-FCN holds an upper hand. When the GPU time is increased, AlexNET, with the aid of Resnet, the feature extraction has a greater Map than when compared to the inception of the Resnet extraction model. The proposed Resnet 101- Alexnet gives a higher mAP value of 54% for GPU time of 1000 milliseconds, whereas other methods such as Inception Resnet-SSD, Inception Resnet-R-FCN, Inception Resnet-RCNN, Inception Resnet-Alexnet, Resnet 101-SSD, Resnet 101-R-FCN, and Resnet 101-RCNN give mAP values of 35%, 37%,39%, 48%, 40%, 42% and 44% respectively.

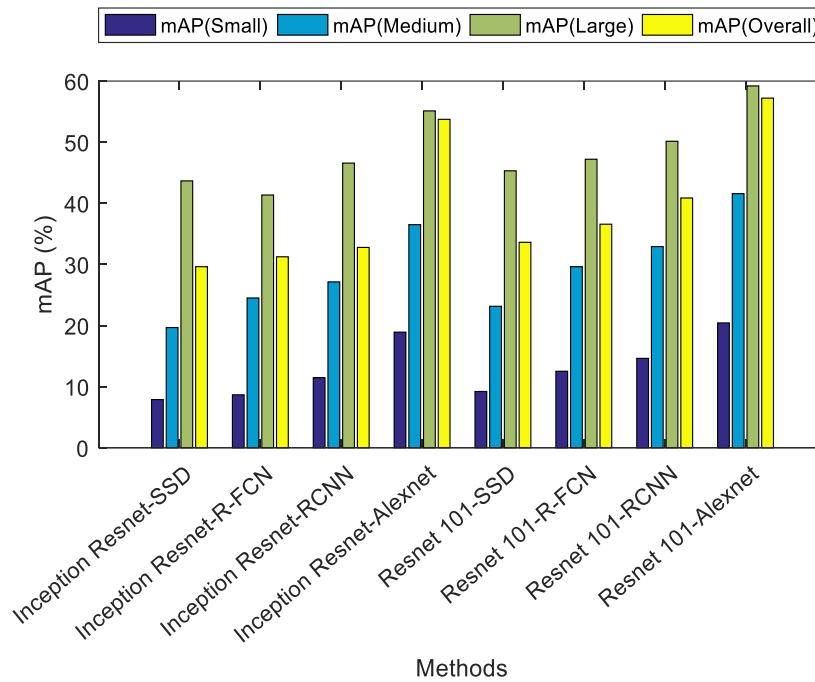


Figure 6: Accuracy stratified by object size, Meta-Architecture, and Feature extractor, Image resolution to 300

In Figure 6, the performance analysis is done on the different models with various sizes of objects. It is not surprising to see that all the methods perform better in the case of large objects. It can be concluded that the SSD techniques have meager performance in case of small objects and are competitive with faster RCNN and R-FCN in case of large objects, which outperforms these meta-architectures for faster and lightweight extractors. The proposed Resnet 101- Alexnet gives a higher mAP value (Large) of 59.21%, whereas other methods such as Inception Resnet-SSD, Inception Resnet-R-FCN, Inception Resnet-RCNN, Inception Resnet-Alexnet, Resnet 101-SSD, Resnet 101-R-FCN, and Resnet 101-RCNN give mAP values of 43.67%, 41.36%,46.58%, 55.12%, 45.32%, 47.21% and 50.15% respectively.

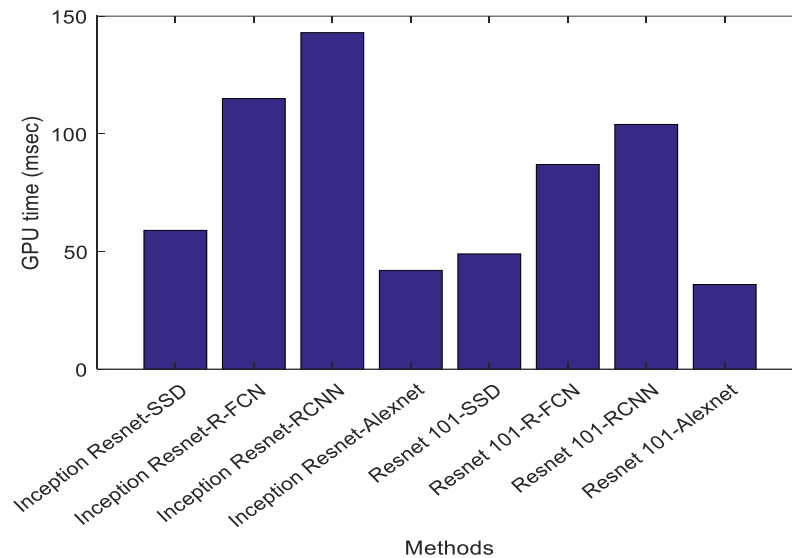


Figure 7: GPU Time (Milliseconds) for each model, for image resolution of 300

In Figure 7, a plot of GPU time for each of the models is shown. This has a constraint of platform dependency. The counting and floating-point operations provide us a platform-independent measure for the computation, which shall or shall not be linear with respect to the exact running cost and times owing to the number of issues such as caching, I/O, and optimization, etc. The GPU time of the proposed Alexnet and the Inception Resnet has a faster GPU time of the order of 42 and 36 Milliseconds.

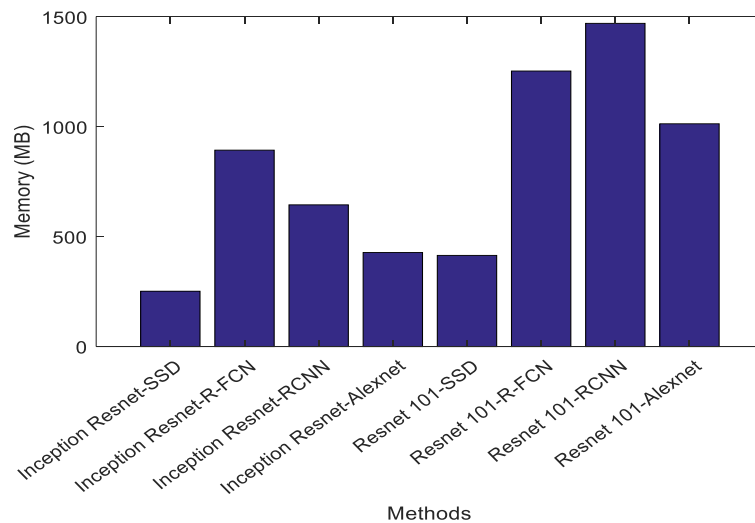


Figure 8: Memory (Mb) usage for each model vs. feature extraction methods

Figure 8 measures the total usage of memory other than the peak memory usage. These error bars reflect the variance in the usage of memory by the various proposals for faster R-CNN. The figure plots a few of the similar information in a more detailed manner by drilling down into the meta-architecture and the feature extractor selections. It can be concluded from the figure that the proposed Alexnet with the Resnet 101 and the incnet has a lower memory usage of the order of 1014

MB and 428 MB, which has lesser usage when these are compared to the R-FCN and RCNN, respectively.

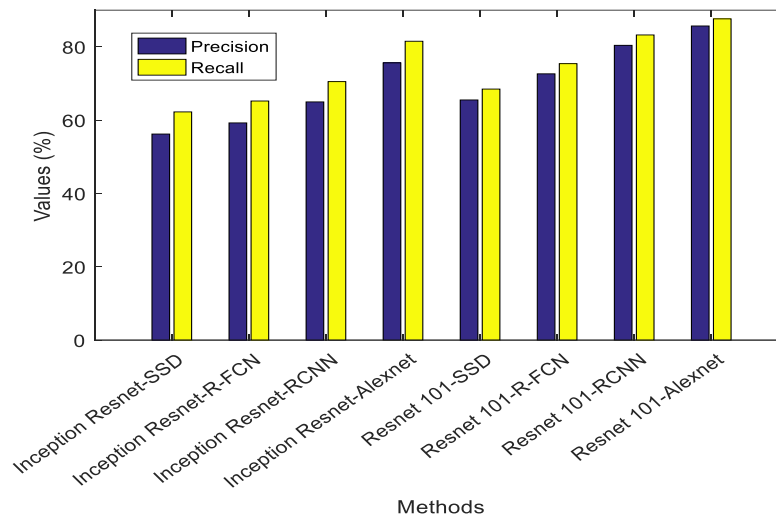


Figure 9: Precision and Recall results comparison of each model vs. feature extraction methods

Figure 9 shows the precision and recall comparison by using different numbers of proposals. From the figure, it is concluded that the proposed Alexnet with Resnet 101 and Inception Resnet has a higher precision value of 85.7143% and 75.63% whereas other methods, such as Inception Resnet-SSD, Inception Resnet-R-FCN, Inception Resnet-RCNN, Restnet 101-SSD, Restnet 101-R-FCN, Restnet 101-RCNN, have given precision values of 56.21%, 59.25%, 64.98%, 65.51%, 72.63% and 80.41% respectively.

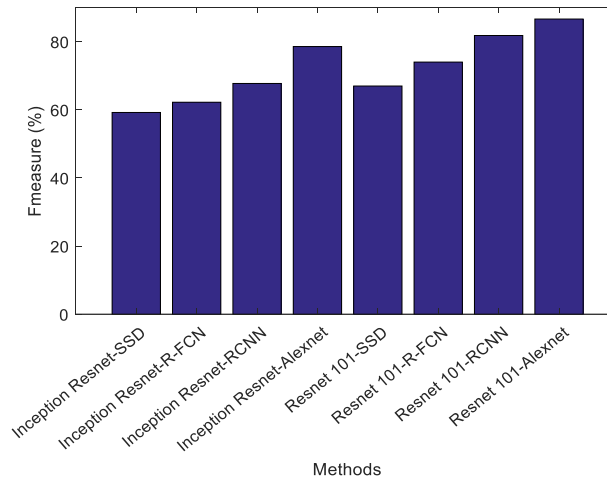


Figure 10: F- measure results comparison of each model vs. feature extraction methods

Figure 10 shows the f-measure comparison by using different numbers of proposals. From the figure, it is concluded that the proposed Alexnet with Resnet 101 and Inception Resnet has a higher f-measure value of 87.2727% and 78.575% whereas other methods, such as Inception Resnet-SSD, Inception Resnet-R-FCN, Inception Resnet-RCNN, Restnet 101-SSD, Restnet 101-R-FCN, Restnet 101-

RCNN, have given f-measure values of 59.23%, 62.235%, 67.745%, 66.995%, 74.02% and 81.825% respectively.

5 Conclusion and Future Work

Object detection is one of the important and widely used methods for the identification of target objects in a specific image and for the determination of the position and category for achieving machine vision's understanding. In this research, the dimensionality reduction is addressed using the BEMD technique for achieving high-order object detection that even in a low-dimensional space of dimension. The identification of the extrema in a two-dimensional image is carried out through searching for extreme values in eight different directions individually among various columns, rows, and diagonals as tabulated pixels. For each object detection, the methodology which are introduced is to minimize the combined effect of classification and the regression loss. Implementation of flexible and unified implementations are of four meta-architectures, R-FCNs s R-CNN, and Alexnet classifiers. AlexNet is designed in a deeper way than the standard CNN, with five CNNs followed by three pooling layers. The COCO detection is taken as a challenge that provides bounding box coordinates of more than 200,000 images with 80 object categories. This data is used for the comparison of results with other object detection methods. The performance metrics used are the Average precision (AP), Mean Average precision (mAP), the precision, and the recall with F-Measure. The results indicate that when fewer proposals are used, the Alexnet can speed up very fast without involving huge losses in terms of accuracy, and thus making it competitive with a faster R-FCN, SSD, and the RFCN. The proposed way of implementation does not require any modifications of the detection models for matching with different input formats and parameters that tune to object detection, and the same is left as the scope for future study.

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